



IILM UNIVERSITY, GREATER NOIDA
School of Computer Science and Engineering

Internship Report

Java Full Stack Development

Submitted in partial fulfilment of the requirement for the degree of B.Tech CSE

Anuj Singh | Roll No: CS-2341597



Internship Credential

CERTIFICATE OF VIRTUAL INTERNSHIP

Presented to Anuj Singh for successfully completing the 8-week Java Full Stack Developer Virtual Internship, supported by EduSkills Academy under AICTE.

- **Program** Java Full Stack Developer
- **Duration** 8 Weeks (June –August 2026)
- **Certificate ID** 4554be59aaa35039b629
- **Student ID** STU69971eC2b4f831771511490
- **Grade** E (Excellent, 80–89)

CANDIDATE'S DECLARATION

I, Anuj Singh, declare that this internship report titled “Java Full Stack Development” has been completed by me for the award of B.Tech CSE. All references have been quoted and no material has been taken from any other source without acknowledgement.

Anuj Singh
Roll No: CS-2341597

Introduction

An eight-week learning program covering the complete path from building the visible part of a web page to developing server-side logic and connecting applications with a relational database.

Main objective: understand how a UI communicates with application logic, how application logic interacts with a database, and how source code is maintained through version control.

The internship connected concepts often studied separately in college —structure, styling, interactivity, programming logic, persistence, and data storage —into one coherent development workflow.

8

Weeksofstructured learning

10

Technologiesacross the stack

4

Architectural layers connected

Problem Statement & Objectives

How can a developer design a web application where the front end provides an interactive experience, the Java-based backend handles business logic, the database stores and retrieves information efficiently —and changes are managed safely through version control?

- 1 Understand web page structure using HTML
- 2 Learn CSS fundamentals & responsive design with Bootstrap
- 3 Handle client-side interaction with JavaScript & jQuery
- 4 Strengthen programming with Core Java
- 5 Explore enterprise concepts in Advanced Java
- 6 Learn Spring for Java-based enterprise applications
- 7 See how Spring Boot simplifies development
- 8 Apply Hibernate & Object-Relational Mapping
- 9 Use MySQL for structured data storage
- 10 Practice source-code management with Git

Scope of the Project

- | | | |
|----------------------|--|---|
| ● Front End | HTML, CSS, Bootstrap, JavaScript, jQuery | Page structure, styling, responsiveness & interaction |
| ● Programming | Core Java, Advanced Java | Foundation for application development |
| ● Back End | Spring, Spring Boot | Java-based enterprise & application development |
| ● Data Layer | Hibernate, MySQL | ORM concepts & relational data storage |
| ● Dev Process | Git & Version Control | Tracking and managing source-code changes |

Technologies & Tools Used



HTML



CSS



Bootstrap



JavaScript



jQuery



Core Java



Advanced Java



Spring



Spring Boot



Hibernate



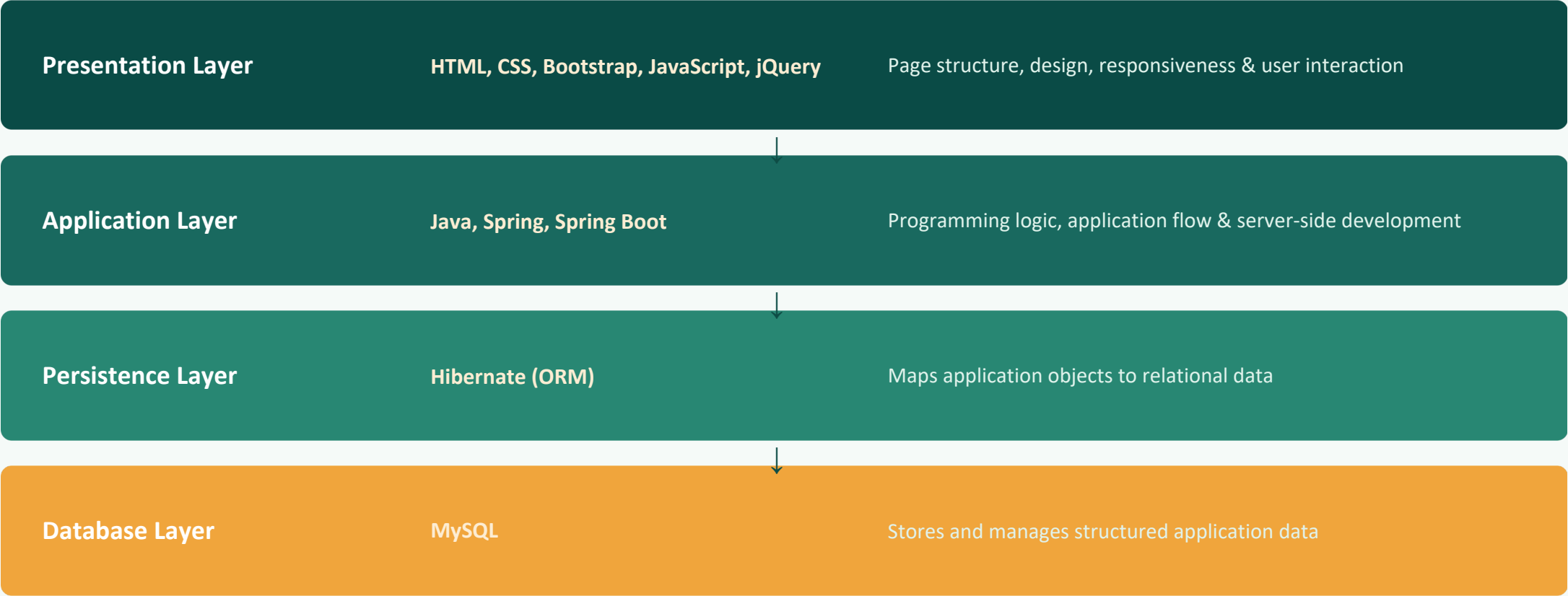
MySQL



Git

System Architecture

Conceptual layered architecture of the Java Full Stack learning environment



Methodology

A progressive, layer-by-layer learning path

Web Fundamentals

HTML, CSS, Bootstrap

Java Programming

Core Java

Spring & Spring Boot

Framework & simplified config

Version Control

Git



Client-Side Interaction

JavaScript, jQuery

Enterprise Java

Advanced Java

Database Integration

Hibernate, MySQL

Workflow: Understand concept → Study syntax/structure → Apply through exercises → Debug → Connect with next layer → Review → Track changes with version control.

Challenges & How I Approached Them

1 Understanding the Full Stack

Learned the stack layer by layer instead of trying to grasp everything at once.

2 Connecting Java with Databases

Focused on the complete data flow from application to persistence layer to MySQL.

3 Learning Frameworks

Treated Spring & Spring Boot as extensions of Java rather than unrelated tools.

4 Debugging

Read the error, identified the affected layer, checked the code, then tested the change.

Expected Outcomes



Explain the basic role of front-end technologies



Improved understanding of responsive design (CSS, Bootstrap)



Confidence in JavaScript & jQuery for client-side interaction



Stronger foundation in Core Java programming



Basic understanding of enterprise Java application development



Understanding of Spring & Spring Boot



Grasp of ORM concepts through Hibernate



Basic understanding of relational data storage using MySQL



Awareness of Git & version control importance



Clearer picture of how front-end, back-end & database work together

Personal Reflection

The internship gave a realistic view of full-stack development —designing interfaces, handling data, writing logic, working with databases, testing, and maintaining source code. Breaking the stack into smaller parts made a large set of technologies manageable, and debugging became a normal, expected part of learning.

Conclusion

This 8-week internship connected front-end design, Java programming, enterprise frameworks, database integration, and version control into one coherent full-stack development workflow.

Looking ahead: continuing to strengthen Java, Spring Boot, database development, problem solving, and software engineering practices for larger real-world projects.

KEY TAKEAWAY

Frameworks are not replacements for fundamentals
—Core Java and relational database concepts make Spring, Spring Boot, and Hibernate easier to understand.

Consistency over intensity: breaking a wide stack into manageable phases made deep learning possible.



Thank You

Java Full Stack Development Internship — IILM University

Anuj Singh | Roll No: CS-2341597 | B.Tech CSE